

Homework 6.12

Problem 1

Using the notation of Theorem 3 and Corollary 1, show that if an exponential polynomial $g(x) = \sum_{k=0}^{N-1} a_k E_k(x)$ assumes the value 0 at each node x_j , then the coefficients a_k are all 0.

Note:

$$\begin{aligned}
 a_k &= \langle f, E_k \rangle = \frac{1}{N} \sum_{j=0}^{N-1} f(x_j) \overline{E_k(x_j)} \\
 \Rightarrow g(x_v) &= 0 = \sum_{k=0}^{N-1} \left[\frac{1}{N} \sum_{j=0}^{N-1} f(x_j) \overline{E_k(x_j)} \right] E_k(x_v) \\
 \Rightarrow 0 &= \sum_{j=0}^{N-1} f(x_j) \left[\frac{1}{N} \sum_{k=0}^{N-1} \overline{E_j(x_k)} E_v(x_k) \right] = \sum_{j=0}^{N-1} f(x_j) \langle E_j, E_v \rangle = f(x_k) \\
 \Rightarrow a_k &= \langle f, E_k \rangle = \frac{1}{N} \sum_{j=0}^{N-1} f(x_j) \overline{E_k(x_j)} = \frac{1}{N} \sum_{j=0}^{N-1} (0) * \overline{E_k(x_j)} = 0 \\
 &\therefore a_k = 0
 \end{aligned}$$

Problem 6

Show that the inner product defined in Equation (12) satisfies the three Properties **1, 2, and 3** following the equation.

$$\langle f, g \rangle_N = \frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \overline{g(x_k)}, x_k = \frac{2\pi j}{N}$$

$$1) \langle f, f \rangle_N \geq 0$$

$$\langle f, f \rangle_N = \frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \overline{f(x_k)} = \frac{1}{N} \sum_{k=0}^{N-1} |f(x_k)|^2$$

$$|f(x_k)|^2 \geq 0 \Rightarrow \frac{1}{N} \sum_{k=0}^{N-1} |f(x_k)|^2 \geq 0$$

$$2) \langle f, g \rangle_N = \overline{\langle g, f \rangle_N}$$

$$\langle f, g \rangle_N = \frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \overline{g(x_k)} = \frac{1}{N} \sum_{k=0}^{N-1} \overline{g(x_k)} f(x_k) = \overline{\frac{1}{N} \sum_{k=0}^{N-1} g(x_k) \overline{f(x_k)}} = \overline{\langle g, f \rangle_N}$$

$$3) \langle \alpha f + \beta g, h \rangle_N = \alpha \langle f, h \rangle_N + \beta \langle g, h \rangle_N$$

$$\begin{aligned}\langle \alpha f + \beta g, h \rangle_N &= \frac{1}{N} \sum_{k=0}^{N-1} [\alpha f(x_k) + \beta g(x_k)] \overline{h(x_k)} = \frac{1}{N} \sum_{k=0}^{N-1} [\alpha f(x_k) h(x_k) + \beta g(x_k) h(x_k)] \\ \langle \alpha f + \beta g, h \rangle_N &= \alpha \frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \overline{h(x_k)} + \beta \frac{1}{N} \sum_{k=0}^{N-1} g(x_k) \overline{h(x_k)} = \alpha \langle f, h \rangle_N + \beta \langle g, h \rangle_N \\ \therefore \langle \alpha f + \beta g, h \rangle_N &= \alpha \langle f, h \rangle_N + \beta \langle g, h \rangle_N\end{aligned}$$

Why is $\| \cdot \|$ not a norm?

For something to be a norm, we require that $\|f\| = 0 \Leftrightarrow f = 0$, But take $f(x) = \sin(Nx)$

$$\|f\| = (\langle f, f \rangle)^{\frac{1}{2}} = \left(\frac{1}{N} \sum_{k=0}^{N-1} f(x_k) \overline{f(x_k)} \right)^{\frac{1}{2}} = \left(\frac{1}{N} \sum_{k=0}^{N-1} (0 * 0) \right)^{\frac{1}{2}} = 0 \nRightarrow f = 0$$

Homework 6.13

Problem 2

Show that f and $f + \lambda E_k$ are indistinguishable if they are interpolated at points $x_j = \frac{2\pi j}{N}$, with k a multiple of N . Thus, to detect all components of f having frequency $\leq \omega$, the sampling frequency should be greater than ω . This is another aspect of the aliasing effect.

$$\begin{aligned}g(x) &= f(x) + \lambda E_k(x) \\ p_f(x) &= \frac{1}{N} \sum_{l=0}^{N-1} \langle f, E_l \rangle_N E_l, \quad p_g(x) = \frac{1}{N} \sum_{l=0}^{N-1} \langle g, E_l \rangle_N E_l \\ \langle f, E_l \rangle_N &= \frac{1}{N} \sum_{j=0}^{N-1} f(x_j) \overline{E_l(x_j)} \\ \langle g, E_l \rangle_N &= \frac{1}{N} \sum_{j=0}^{N-1} g(x_j) \overline{E_l(x_j)} = \frac{1}{N} \sum_{j=0}^{N-1} [f(x_j) + \lambda E_k(x_j)] \overline{E_l(x_j)} \\ \Rightarrow \langle g, E_l \rangle_N &= \frac{1}{N} \sum_{j=0}^{N-1} [f(x_j) \overline{E_l(x_j)} + \lambda E_k(x_j) \overline{E_l(x_j)}] = p_f(x) + \lambda \frac{1}{N} \sum_{j=0}^{N-1} E_k(x_j) \overline{E_l(x_j)} \\ &\Rightarrow \langle g, E_l \rangle_N = p_f(x) + \lambda \langle E_k, E_l \rangle_N = p_f(x) + \lambda \\ \therefore p_g(x) &= p_f(x) + \lambda\end{aligned}$$

Note that the only difference is that of a shift in the graph.

Problem 3

(Aliasing) Two functions can appear to be the same if we know only samples of the functions taken at discrete points. Show that these two functions appear to be identical when sampled at the integer values of x .

$$f(x) = \cos[(n - \alpha)\pi x + \beta], \quad g(x) = \cos[(n + \alpha)\pi x - \beta]$$

$$p_f = \sum_{k=0}^{N-1} \langle f, E_k \rangle_N E_k(x), \quad p_g = \sum_{k=0}^{N-1} \langle g, E_k \rangle_N E_k(x)$$

To show that $p_f(x_z) = p_g(x_z)$, $x_z \in \mathbb{Z}$, it is enough to show that at integer values of x , the functions become the same.

$$\begin{aligned} P_f &= \sum_{k=0}^{N-1} \langle \cos[(n - \alpha)\pi x + \beta], E_k \rangle_N E_k(x) \\ \Rightarrow p_f &= \sum_{k=0}^{N-1} \langle \cos[(n - \alpha)\pi x] \cos(\beta) - \sin[(n - \alpha)\pi x] \sin(\beta), E_k \rangle_N E_k(x) \\ \Rightarrow p_f &= \sum_{k=0}^{N-1} \langle \cos(\beta) [\cos(n\pi x) \cos(a\pi x) - \sin(n\pi x) \sin(a\pi x)] - \sin(\beta) [\sin(n\pi x) \cos(a\pi x) \\ &\quad - \cos(n\pi x) \sin(a\pi x)], E_k \rangle_N E_k(x) \\ &\quad \sin(n\pi x) = 0, x \in \mathbb{Z} \\ \Rightarrow p_f &= \sum_{k=0}^{N-1} \langle \cos(\beta) [\cos(n\pi x) \cos(a\pi x)] + \sin(\beta) [\cos(n\pi x) \sin(a\pi x)], E_k \rangle_N E_k(x) \\ P_g &= \sum_{k=0}^{N-1} \langle \cos[(n + \alpha)\pi x - \beta], E_k \rangle_N E_k(x) \\ \Rightarrow p_g &= \sum_{k=0}^{N-1} \langle \cos[(n + \alpha)\pi x] \cos(\beta) + \sin[(n - \alpha)\pi x] \sin(\beta), E_k \rangle_N E_k(x) \\ \Rightarrow p_g &= \sum_{k=0}^{N-1} \langle \cos(\beta) [\cos(n\pi x) \cos(a\pi x) - \sin(n\pi x) \sin(a\pi x)] + \sin(\beta) [\sin(n\pi x) \cos(a\pi x) \\ &\quad + \cos(n\pi x) \sin(a\pi x)], E_k \rangle_N E_k(x) \\ &\quad \sin(n\pi x) = 0, x \in \mathbb{Z} \\ \Rightarrow p_g &= \sum_{k=0}^{N-1} \langle \cos(\beta) [\cos(n\pi x) \cos(a\pi x)] + \sin(\beta) [\cos(n\pi x) \sin(a\pi x)], E_k \rangle_N E_k(x) \\ &\quad \therefore p_f = p_g \end{aligned}$$

Problem 7

This problem and the next three give a matrix formulation of the fast Fourier transform. Fix an index n and a function f . Let u and v be column vectors having components

$$u_j = f\left(\frac{2\pi j}{n}\right), \quad v_j = \langle f, E_j \rangle_n, \quad (0 \leq j \leq n-1)$$

Prove that $v = Au$, where A is the matrix with components $A_{jk} = n^{-1} \exp\left[-\frac{2\pi ijk}{n}\right]$, $(0 \leq j, k \leq n-1)$.

NTS:

$$A_j u = v_j$$

$$A_j = [A_{jk}], \quad 0 \leq k \leq n-1$$

$$A_j = \frac{1}{n} \begin{bmatrix} 1 & \exp\left[-\frac{2\pi i j}{n}\right] & \dots & \exp\left[-\frac{2\pi i j(n-1)}{n}\right] \end{bmatrix}$$

$$A_j u = \left(\frac{1}{n} \begin{bmatrix} 1 & \exp\left[-\frac{2\pi i j}{n}\right] & \dots & \exp\left[-\frac{2\pi i j(n-1)}{n}\right] \end{bmatrix} \right) \cdot \begin{bmatrix} f(0) \\ f\left(\frac{2\pi}{n}\right) \\ \vdots \\ f\left(\frac{2\pi(n-1)}{n}\right) \end{bmatrix}$$

$$A_j u = \frac{1}{n} \left(f(0) + f\left(\frac{2\pi}{n}\right) \exp\left[-\frac{2\pi i j}{n}\right] + \dots + f\left(\frac{2\pi(n-1)}{n}\right) \exp\left[-\frac{2\pi i j(n-1)}{n}\right] \right)$$

$$\Rightarrow A_j u = \frac{1}{n} \left(f(0) E_j(0) + f\left(\frac{2\pi}{n}\right) E_j\left(-\frac{2\pi}{n}\right) + \dots + f\left(\frac{2\pi(n-1)}{n}\right) E_j\left(-\frac{2\pi(n-1)}{n}\right) \right)$$

$$\Rightarrow A_j u = \frac{1}{n} \left(f(0) \overline{E_j(0)} + f\left(\frac{2\pi}{n}\right) \overline{E_j\left(\frac{2\pi}{n}\right)} + \dots + f\left(\frac{2\pi(n-1)}{n}\right) \overline{E_j\left(\frac{2\pi(n-1)}{n}\right)} \right)$$

$$\Rightarrow A_j u = \frac{1}{n} \sum_{l=0}^{n-1} f(x_l) \overline{E_j(x_l)} = \langle f, E_j \rangle_n = v_j$$

Since we have that j is arbitrary:

$$\therefore Au = v$$

Problem 8

(Continuation) Prove that the matrix A contains only n different elements.

Note:

$$n \equiv 0 \pmod{n}$$

Since jk produces numbers from 0 to $(n-1)^{n-1}$ it seems that we would have that a lot of entries but remember that since $e^0 = e^{\frac{2\pi n}{n}}$ we have that we are in mod n arithmetic so we only really have n inique entries.

Prove that $n^{\frac{1}{2}}A$ is a unitary matrix.

$$\begin{aligned} \left(n^{\frac{1}{2}}A\right)\left(n^{\frac{1}{2}}A\right)^* &\Rightarrow \left[\left(n^{\frac{1}{2}}A\right)\left(n^{\frac{1}{2}}A\right)^*\right]_{jk} = n \sum_{p=0}^{n-1} A_{jp} A_{pk}^* = n \sum_{p=0}^{n-1} A_{jp} \overline{A_{kp}} \\ \left[\left(n^{\frac{1}{2}}A\right)\left(n^{\frac{1}{2}}A\right)^*\right]_{jk} &= n \sum_{p=0}^{n-1} \left[n^{-1} \exp\left[-\frac{2\pi ijp}{n}\right]\right] \left[n^{-1} \exp\left[\frac{2\pi ikp}{n}\right]\right] \\ \left[\left(n^{\frac{1}{2}}A\right)\left(n^{\frac{1}{2}}A\right)^*\right]_{jk} &= \frac{1}{n} \sum_{p=0}^{n-1} \left[\exp\left[\frac{2\pi ikp}{n}\right] \exp\left[-\frac{2\pi ijp}{n}\right]\right] = \frac{1}{n} \sum_{p=0}^{n-1} E_k(x_p) \overline{E_j(x_p)} = \langle E_k, E_j \rangle = \delta_{jk} \\ &\therefore \left(n^{\frac{1}{2}}A\right)\left(n^{\frac{1}{2}}A\right)^* = I \end{aligned}$$

Prove that A is symmetric.

$$A = \left[n^{-1} \exp\left[-\frac{2\pi ijk}{n}\right]\right] = \left[n^{-1} \exp\left[-\frac{2\pi ikj}{n}\right]\right] = A^T$$